



PiCloud

An Easy Way to the Cloud

This article provides an introduction to PiCloud, a cloud computing solution.

There are three components to PiCloud:

1. *PiCloud* refers to the cloud computing infrastructure provided by PiCloud Inc.
2. *Client* refers to the computer or program that is interacting with PiCloud, i.e., submitting jobs, retrieving results, etc.
3. *cloud* refers to the Python library used to interact with the cloud.

Installation

Although primarily a commercial offering, PiCloud has five free compute hours as part of their *Free Developer Account*, which should be good enough to get a feel of working with it. The first step is to register at <http://www.picloud.com>, which will require you to confirm your email account.

The next step is to install the Python client, which is most easily done with *pip* (<http://pypi.python.org/pypi/pip>), i.e., `sudo pip install cloud` will do the installation for you (it currently supports Python 2.5, 2.6 and 2.7). Next, you want to authenticate your computer to use PiCloud, where you have to enter your PiCloud login information provided in the following registration step:

```
$ sudo picloud setup
```

```
Please enter your PiCloud account login information.
```

```
If you do not have an account, please create one at http://www.
```

```
picloud.com
```

```
E-mail: echorand@gmail.com
```

```
Password:
```

PiCloud uses API keys, rather than your login information, to authenticate your machine. In the event your machine is compromised, you can deactivate your API key to disable access. In this next step, you can choose to use an existing API key for this machine, or create a new one. We recommend that each machine has its own unique key.

```
Your API Key(s)
```

```
3241
```

```
Please select an API Key or just press enter to create a new one automatically:
```

```
API Key: 3242
```

Please note that your API key may be different. You can see the API keys you have registered in the Web console when you log in to your account at <http://www.picloud.com/>, where you will also find a secret key corresponding to your API key. You will need to make this information available to PiCloud when you try to submit your job. One way of doing this is setting it in the file, `$HOME/.picloud/cloudconf.py`. For